Kernels and Gaussian Processes* 12

We will introduce here kernels and apply the concept of duality to Bayesian regression. In this setting we will use kernels inside probabilistic models for regression and classification tasks, and this will lead us to the framework of Gaussian Processes, which are one of the most flexible approaches for supervised learning, especially when we have a limited number of observations.

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12.1 Equivalent Kernel

Consider a predictive distribution on a Bayesian Linear Regression model

$$p(y|x, \underline{x}, \underline{y}, \alpha, \beta) = \mathcal{N}\left(y|m_N^T\phi(x), \sigma_N^2(x)\right)$$

Let's focus our attention on the mean predictor, by rewriting the expression as

$$y(x, m_N) = m_N^T\phi(x) = \beta\phi(x)^T S_N \Phi \underline{y} = \sum_{n=1}^{N} \beta\phi(x)^T S_N \phi(x_N) y_n = \sum_{n=1}^{N} k(x, x_n) y_n$$

Here we can observe that the predictive mean is a linear combination of the observations that we have weighted by a term, which is defined, in our case, as:

$$k(x, x') = \beta\phi^T(x) S_N \phi(x')$$

and it is called the **equivalent kernel**.

It acts as a sort of smoother that depends on the distance between x and x_n.

If we evaluate the covariance in between

$$cov[y(x), y(x')] = \phi^T(x) S_N \phi(x') = \beta^{-1} k(x, x')$$

Notice also that we can rewrite the equivalent kernel as

$$k(x, x') = \Psi^T(x)\Psi(x) \qquad \Psi(x) = \beta^{\frac{1}{2}} S_N^{\frac{1}{2}} \phi(x)$$

I.e. we can see the equivalent kernel as a rescaled scalar product in between our basis functions.

12.2 Dual Formulation of Linear Regression

Let's restate our problem. We have observations $\underline{x}, \underline{y} = (x_n, y_N)_{n \in 1,\dots,N}$ and a linear model $w^T\phi(X)$. Whenever we optimize the weights, we are evaluating our basis function on the input points. Which means that the

solution that we are searching is contained in the subspace spanned by $< \phi(x_1), \ldots, \phi(x_n) >$. Which means that we can rewrite our vector of weight as

$$w = \sum_{j=1}^{N} a_j \phi(x_j)$$

So basically we can use variables a instead of variables w. Variables a are known as **dual variables**. Now let's introduce a kernel in between two observations points $k(x_i, x_j) = \phi^T(x_i)\phi(x_j)$ and the corresponding **Gram Matrix**, i.e.

$$K = (k_{ij}) \qquad k_{ij} = k(x_i, x_j) \qquad K^i i^{th} \text{column}$$

which means that we have

$$w^T \phi(x_i) = a^T K^i$$

If we write the regularized problem that we are trying to solve, then we can write

$$E_D(a) + \lambda E_w(a) = \sum_{n=1}^{N} (y_n - a^T K^n)^2 + \lambda a^T K a$$

We still have a convex optimization problem, which means that we can compute the maximum likelihood solution, which is

$$\hat{a} = (k + \lambda I)^{-1} \underline{y} \qquad \text{ML solution}$$

whereas the prediction on a new point $x*$ would be

$$y(x*) = K_*^T (K + \lambda I)^{-1} \underline{y}$$
$$K_* = (k(x_*, x_1), \ldots, k(x_*, x_n))$$

This new formulation is what is known as the **Kernel trick**: we don't need to know the basis function, we just need their scalar product (which can allow us also to work with infinite dimensional function spaces). Moreover, the kernel trick formulation has a complexity of the order $O(N^3)$, therefore it depends on the number of observations, whilst the basis function formulation depends on the number of basis functions considered $O(M^3)$.

12.3 Random Functions

A **Random Function** can be thought of as a probability distribution over functions. More formally:

Definition 12.3.1 *A random function (or stochastic process) is an infinite collection of random variables, indexed by the argument of the function.*

In order to have an intuition of what this means, consider a vector $\mathbf{x} \in \mathbb{R}^n$ and pick a function $f(\mathbf{x}) = w_0\phi_0(\mathbf{x}) + \ldots + w_{M-1}\phi_{M-1}(\mathbf{x})$ (all $\phi_i(\mathbf{x})$ are fixed).

Consider the set $\mathcal{F} = \{f(\mathbf{x}) = w_0\phi_0(\mathbf{x}) + \ldots + w_{M-1}\phi_{M-1}(\mathbf{x})\}$, if all ϕ_i $i \in 0, \ldots, M-1$ are orthogonal, it can be easily proved that $\mathcal{F}$ is a M-dimensional vector space, where $\Phi(\mathbf{x}) = (\phi_j(\mathbf{x}))$ is the set of basis functions.

We want to consider a distribution over the function space $\mathcal{F}$: observing that each function in $\mathcal{F}$ is identified by its vector of coefficients $\mathbf{w}$, then a distribution over $\mathcal{F}$ is equivalent to a distribution over $\mathbf{w}$.

Under certain conditions (i.e. when $\mathcal{F}$ is a Polish space) the distribution over $\mathcal{F}$ is determined by **finite dimensional projections**.

That is, a distribution is practically described by knowing: $p(f(\mathbf{x}_1), \ldots, f(\mathbf{x}_N))$, $\forall \mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^n$.

Consider a Gaussian prior over the coefficients $\mathbf{w} \sim \mathcal{N}(0, \mathbb{I})$.

If we fix a single point $\mathbf{x} \in \mathbb{R}^n$, i.e. we look at the 1-dim marginal $p(f(\mathbf{x}))$, since $f(\mathbf{x}) = \sum_{j=0}^{M-1} w_j\phi_j(\mathbf{x}) \in \mathbb{R}$ with $w_j \sim \mathcal{N}(0,1)$, then $f(\mathbf{x}) \sim \mathcal{N}(0, \phi^T\phi)$ by the properties of the Gaussian distribution.

Consider now 2-dim marginals, that is take $f \in \mathcal{F}$ and fix two points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, our goal is to study $p(f(\mathbf{x}_1), f(\mathbf{x}_2))$, i.e. find its mean and covariance matrix (notice that $f(\mathbf{x}_1), f(\mathbf{x}_2)$ are linear combinations of Gaussians, so this is again a Gaussian distribution). It turns out that:

$$p(f(\mathbf{x}_1), f(\mathbf{x}_2)) = \mathcal{N}\left(0, \begin{pmatrix}\Phi(\mathbf{x}_1)\\ \Phi(\mathbf{x}_2)\end{pmatrix}^T \mathbb{I} \begin{pmatrix}\Phi(\mathbf{x}_1)\\ \Phi(\mathbf{x}_2)\end{pmatrix}\right)$$

That is:

$$cov(f(\mathbf{x}_1), f(\mathbf{x}_2)) = \begin{pmatrix}\Phi^T(\mathbf{x}_1)\Phi(\mathbf{x}_1) & \Phi^T(\mathbf{x}_1)\Phi(\mathbf{x}_2)\\ \Phi^T(\mathbf{x}_2)\Phi(\mathbf{x}_1) & \Phi^T(\mathbf{x}_2)\Phi(\mathbf{x}_2)\end{pmatrix} := \begin{pmatrix}k(\mathbf{x}_1, \mathbf{x}_1) & k(\mathbf{x}_1, \mathbf{x}_2)\\ k(\mathbf{x}_1, \mathbf{x}_2) & k(\mathbf{x}_2, \mathbf{x}_2)\end{pmatrix}$$

again using properties of the multivariate Gaussian distribution. Note that entries of the covariance matrix are the evaluations of the kernel in each pair of points.

Analogously, the n-dimensional distribution $p(f(\mathbf{x}_1), \ldots, f(\mathbf{x}_n))$ is going to be $\mathcal{N}(0, K)$, with $K = (K_{ij})$ and $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T\phi(\mathbf{x}_j)$ (i.e. the kernel is the scalar product of the basis functions).

Thus, in the example that we have considered, all finite dimensional projections are Gaussian distributions, and this will be a key point in the definition of Gaussian Processes.

12.4 Gaussian Processes - basic definitions

Definition 12.4.1 *A **Gaussian Process** (GP) is a stochastic process indexed by a continuous variable $\boldsymbol{x} \in \mathbb{R}^n$, such that all finite dimensional distributions*

are multivariate Gaussian.

So a GP f is essentially a collection of random variables $f(\mathbf{x}), \forall \mathbf{x} \in \mathbb{R}^n$ (i.e. uncountably many r.v.), and we are interested in the joint distribution of these random variables.

By definition: $\forall \mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^n, p(f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)) = \mathcal{N}(m_N, \Sigma_N)$

Hence a GP is uniquely defined by a function $\mu : \mathbb{R}^n \to \mathbb{R}$ which computes the mean at every point $\mathbf{x}$, and by a function $K : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ which computes the covariance between any two points.

Thus, exploiting the definition: $f \sim GP(\mu, K)$ if and only if:

$$\forall \mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^n,\ p(\mathbf{f}) = p(f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)) = \mathcal{N}(\underline{\mu}, K)$$

being the mean $\underline{\mu} = (\mu(\mathbf{x}_1), \dots, \mu(\mathbf{x}_n))$, and the covariance matrix (also called **Gram matrix**) $K = (K_{ij})$ (with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$).

Of course K has to be symmetric and positive definite $\forall \mathbf{x}_1, \dots, \mathbf{x}_N$.

In what follows our goal will be to do inference. It is important to stress from the beginning that GP are non-parametric models (only fixed hyperparameters enter in the computations), and this makes them a very powerful method.

12.5 GP Regression

12.5.1 Noise-free case

We start by considering regression in the noise-free case, i.e. we assume to observe exactly the values we want to predict.

Suppose to observe the exact value of the GP at points $\mathbf{x} = x_1, \dots, x_N$, with observations $\mathbf{f} = f(x_1), \dots, f(x_N)$.

Consider also test points $\mathbf{x}^\star = x_1^\star, \dots, x_\nu^\star$, with function values $\mathbf{f}^\star = f(x_1^\star), \dots, f(x_\nu^\star)$ unobserved, i.e. to be estimated.

The prior is a $f \sim GP(0, K)$. Inference in this case is particularly simple: it all boils down to (multivariate) Gaussian properties.

First of all we can compute the joint prior distribution of $\mathbf{f}$ and $\mathbf{f}^\star$, that is:

$$p\begin{pmatrix}\mathbf{f}\\ \mathbf{f}^\star\end{pmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K(\mathbf{x},\mathbf{x}) & K(\mathbf{x},\mathbf{x}^\star) \\ K(\mathbf{x}^\star,\mathbf{x}) & K(\mathbf{x}^\star,\mathbf{x}^\star)\end{bmatrix}\right)$$

where the covariance is a block matrix, where each block is a Gram matrix, with $K(\mathbf{x}, \mathbf{x}^\star) = (K_{ij})_{i \in 1,\dots,N} j \in 1, \dots, \mu$ with $K_{ij} = k(x_i, x_j^\star)$.

If f is observed then we can compute the posterior (note that we are conditioning a Gaussian distribution):

$$p(\mathbf{f}^\star|\mathbf{f}) \sim \mathcal{N}\left(K(\mathbf{x}^\star,\mathbf{x})K(\mathbf{x},\mathbf{x})^{-1}\mathbf{f}, K(\mathbf{x}^\star,\mathbf{x}^\star) - K(\mathbf{x}^\star,\mathbf{x})K(\mathbf{x},\mathbf{x})^{-1}K(\mathbf{x},\mathbf{x}^\star)\right)$$

Since this works for any tuple $\mathbf{x}^\star$, the posterior is a GP.

Remark: the mean of the posterior is not constant anymore.

12.5.2 Noisy setting

If we introduce noise, then we observe $\mathbf{y}(\mathbf{x}) = \mathbf{f}(\mathbf{x}) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ (i.e. we have a component depending on the process and a noise term). Hence observations are Gaussian with 0 mean and covariance $cov(\mathbf{y}) = K(\mathbf{x},\mathbf{x}) + \sigma^2\mathbb{I}$.

So the joint prior between observations $\mathbf{y}$ and test points is:

$$p\begin{pmatrix}\mathbf{y}\\ \mathbf{f}^\star\end{pmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K(\mathbf{x},\mathbf{x}) + \sigma^2\mathbb{I} & K(\mathbf{x},\mathbf{x}^\star) \\ K(\mathbf{x}^\star,\mathbf{x}) & K(\mathbf{x}^\star,\mathbf{x}^\star)\end{bmatrix}\right)$$

Conditioning on observations $\mathbf{y}$ we get the posterior (which is again GP):

$$p(\mathbf{f}^\star|\mathbf{y}) \sim \mathcal{N}\left(K(\mathbf{x}^\star,\mathbf{x})\left[K(\mathbf{x},\mathbf{x})+\sigma^2\mathbb{I}\right]^{-1}\mathbf{y}, K(\mathbf{x}^\star,\mathbf{x}^\star)-K(\mathbf{x}^\star,\mathbf{x})\left[K(\mathbf{x},\mathbf{x})+\sigma^2\mathbb{I}\right]^{-1}K(\mathbf{x},\mathbf{x}^\star)\right)$$

The dominant cost of this procedure is computing the inverse $\left[K(\mathbf{x}^\star,\mathbf{x}) + \sigma^2\mathbb{I}\right]^{-1}$, which is an $N \times N$ matrix, hence complexity is $O(N^3)$ (it depends on the amount of observations).

If we want to predict $f(\mathbf{x}^\star)$ for a single point $\mathbf{x}^\star$, it holds that:

$$f(\mathbf{x}^\star) \sim \mathcal{N}(\mathbf{K}(K + \sigma^2\mathbb{I})^{-1}\mathbf{y}, K(\mathbf{x},\mathbf{x}) - \mathbf{K}^T(K + \sigma^2\mathbb{I})^{-1}\mathbf{K})$$

being $\mathbf{K} = (K(\mathbf{x}^\star,\mathbf{x})) = (k(\mathbf{x}^\star, x_1), \dots, k(\mathbf{x}^\star, x_N))$.

Note that the mean can be rewritten as $\mathbb{E}[f(\mathbf{x}^\star)|\mathbf{y}] = \sum_{i=1}^N \alpha_i k(\mathbf{x}^\star, x_i)$ with $\alpha_i = [K(\mathbf{x},\mathbf{x}) + \sigma^2\mathbb{I}]^{-1}\mathbf{y}$, i.e. as a linear combination of the kernels evaluated on the input points.

12.6 Kernel functions and Hilbert spaces

We can see a kernel as defining an integral operator T_k on a space $\mathcal{X}$ with measure μ:

$$(T_k f)(\mathbf{x}) = \int_{\mathbf{x}} k(\mathbf{x},\mathbf{y})f(\mathbf{y})d\mu(\mathbf{y})$$

Note that this is a function of $\mathbf{x}$.

Kernels have the property of being positive semidefinite if they satisfy the following:

$$\int_{\mathcal{X}\times\mathcal{X}} k(\mathbf{x},\mathbf{y})f(\mathbf{x})f(\mathbf{y})d\mu(\mathbf{x})d\mu(\mathbf{y}) \geq 0 \; \forall f \in L_2(\mathcal{X},\mu)$$

or equivalently if the Gram matrix $K = (K_{ij})$ evaluated on any finite collection of points $\{\mathbf{x}_i, i \in 1,\dots,n\}$ is positive (semi)definite.

An eigenfunction ϕ for kernel k with eigenvalue λ is defined as:

$$\int k(\mathbf{x},\mathbf{y})\phi(\mathbf{x})d\mu(\mathbf{x}) = \lambda\phi(\mathbf{y})$$

There can be ∞-many eigenfunctions, they can be ordered w.r.t. decreasing eigenvalues (in analogy with what happens when working with matrices) and they can be chosen orthogonal (i.e. such that $\int \phi_i(\mathbf{x})\phi_j(\mathbf{x})d\mu(\mathbf{x}) = \delta_{ij}$).

It holds the following theorem, that describes how kernels can be decomposed according to eigenfunctions:

Theorem 12.6.1 *(Mercer's theorem)*
Let $(\mathcal{X},\mu)$ be a finite measure space and $k \in L_\infty(\mathcal{X}^2,\mu^2)$ be a kernel such that $T_k : L_2(\mathcal{X},\mu) \to L_2(\mathcal{X},\mu)$ is positive definite. Let $\phi_i \in L_2(\mathcal{X},\mu)$ be the normalized eigenfunctions of T_k associated with the eigenvalues $\lambda_i > 0$. Then:

1. *the eigenvalues $\{\lambda_i\}_{i=1}^\infty$ are absolutely summable;*
2. $$k(\boldsymbol{x},\boldsymbol{x}') = \sum_{i=1}^\infty \lambda_i\phi_i(\boldsymbol{x})\phi_i^\star(\boldsymbol{x}')$$
   holds μ^2 almost everywhere, where the series converges absolutely and uniformly μ^2 almost everywhere.

This brings us to the concept of **Reproducing Kernel Hilbert Space** (RKHS):

Definition 12.6.1 *Let $\mathcal{H}$ be a Hilbert space of real functions f defined on an index set $\mathcal{X}$. Then $\mathcal{X}$ is called a Reproducing Kernel Hilbert Space endowed with an inner product $\langle\cdot,\cdot\rangle_{\mathcal{H}}$ if there exists a function $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ with the following properties:*

1. *for every $\boldsymbol{x}$, $k(\boldsymbol{x},\boldsymbol{x}')$ as a function of $\boldsymbol{x}'$ belongs to $\mathcal{H}$;*
2. *k has the reproducing property $\langle f(\cdot), k(\cdot,\boldsymbol{x})\rangle_{\mathcal{H}} = f(\boldsymbol{x})$*

Note that from the previous definition it follows that as $k(\mathbf{x},\cdot)$ and $k(\mathbf{x}',\cdot)$ are in $\mathcal{H}$, then $\langle k(\mathbf{x},\cdot), k(\mathbf{x}',\cdot)\rangle_{\mathcal{H}} = k(\mathbf{x},\mathbf{x}')$.

Moreover, the following theorem states that the RKHS uniquely determines k and vice versa:

Theorem 12.6.2 *(Moore-Aronszajn theorem)*
Let $\mathcal{X}$ be an index set. Then for every positive definite function $k(\cdot,\cdot)$ on $\mathcal{X}\times\mathcal{X}$ there exists a unique RKHS, and vice versa.

If we take a RKHS and compute the eigenfunctions of a kernel, then we can use them to represent the elements of the RKHS as a linear combination of the eigenfunctions ϕ_j of k : $f(\mathbf{x}) = \sum_j f_j\phi_j(\mathbf{x})$, with coefficients satisfying the smoothness constraint $\sum_j \frac{f_j^2}{\lambda_j} < \infty$.

Moreover it is possible to show that such functions define an Hilbert space $\mathcal{H}$ with inner product $\langle f, g\rangle_{\mathcal{H}} = \sum_j \frac{f_j g_j}{\lambda_j}$.

This Hilbert space is the RKHS corresponding to the kernel k in which:

$$\langle f(\cdot), k(\cdot,\mathbf{x})\rangle_{\mathcal{H}} = \sum_{i=1}^{N} \frac{f_i\lambda_i\phi_i(\mathbf{x})}{\lambda_i} = f(\mathbf{x})$$
$$\langle k(\mathbf{x},\cdot), k(\mathbf{x}',\cdot)\rangle_{\mathcal{H}} = \sum_{i=1}^{N} \frac{f_i\lambda_i\phi_i(\mathbf{x})\lambda_i\phi_i(\mathbf{x}')}{\lambda_i} = k(\mathbf{x},\mathbf{x}')$$

Furthermore, since $||\cdot||_{\mathcal{H}} = \sqrt{\langle\cdot,\cdot\rangle_{\mathcal{H}}}$, the norm of $k(\mathbf{x},\cdot)$ is $k(\mathbf{x},\mathbf{x}) < \infty$ and it belongs to $\mathcal{H}$.

12.6.1 Some examples of kernel functions

There are three important classes of kernel functions, which can be distinguished by looking at the dependence of the kernel $k(\mathbf{x},\mathbf{y})$ on $\mathbf{x}$ and $\mathbf{y}$:

- **Stationary kernel**: it is a function of $\mathbf{x}-\mathbf{y}$ (i.e. invariant to translations);
- **Isotropic kernel**: it is a function of $||\mathbf{x}-\mathbf{y}||$ (i.e. invariant to rigid motions);
- **Dot-product kernel**: it is a function of $\mathbf{x}^T\mathbf{y}$ (i.e. invariant to rotations w.r.t. the origin)

Moreover we can define the notion of **continuity in mean square** of a GP f at $\mathbf{x}$:

Definition 12.6.2 *A GP f is continuous in mean square at $\mathbf{x}$ if, for each $\mathbf{x}_k \to \mathbf{x}$, it holds that $\mathbb{E}[|f(\mathbf{x}_k) - f(\mathbf{x})||^2] \to 0$ (i.e. the convergence of the sequence of inputs is preserved in mean).*

It can be proved that a process is continuous in mean square at $\mathbf{x}$ if and only if the kernel k is continuous at $k(\mathbf{x},\mathbf{x})$. For stationary and isotropic kernels, k must be continuous at 0 (i.e. we only need to check continuity in a single point).

It holds also that if k is $2h$-th differentiable, then f is h-th differentiable in mean square (in particular, if the kernel is smooth, then also the process

is smooth).

The **Gaussian kernel** (or Squared Exponential kernel) is an isotropic kernel defined as:

$$k(\mathbf{x}, \mathbf{y}) = \alpha \exp\left[- \frac{||\mathbf{x} - \mathbf{y}||^2}{\lambda^2} \right]$$

where α is called the *amplitude* and λ is the characteristic *length-scale*. λ is a crucial parameters, which regulates the speed of decay of the correlation between points. The RKHS of this kernel is enjoys the universality property (i.e. it is dense) in the space of continuous functions over a compact set in $\mathbb{R}^n$.

A generalization of this kernel is the **Automatic Relevance Detection Gaussian Kernel**:

$$k(\mathbf{x}, \mathbf{y}) = \alpha \exp\left[- \sum_i \frac{|x_i - y_i|^2}{\lambda_i^2} \right]$$

where there is a different λ_i for each coordinate of the input points.

The **Matérn kernel** is another isotropic kernel defined as:

$$k_{matern}(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{\ell} \right)^{\nu} K_\nu \left(\frac{\sqrt{2\nu} r}{\ell} \right)$$

with $\nu, \ell > 0$ are parameters and K_ν is a modified Bessel function.

The parameter μ regulates the differentiability of the kernel: if $\nu > h$ then the GP with Matérn kernel is h times differentiable (in mean square). For $\nu \to \infty$, the Matérn kernel becomes the Gaussian kernel.

A typical choice for MK is $\nu = p + \frac{1}{2}$:

$$k_{\nu=p+\frac{1}{2}}(r) = \exp\left(- \frac{\sqrt{2\nu} r}{\ell} \right) \frac{\Gamma(p+1)}{\Gamma(2p+1)} \sum_{i=0}^{p} \frac{(p+i)!}{i!(p-i)!} \left(\frac{\sqrt{8\nu} r}{\ell} \right)^{p-i}$$

For $\nu = \frac{1}{2}$ we get the Exponential Kernel:

$$k(\mathbf{x}, \mathbf{y}) = \exp\left[\frac{||\mathbf{x} - \mathbf{y}||}{\lambda} \right]$$

which is continuous but nowhere differentiable.

For what concerns dot-product kernels, we have the **Polynomial kernel**, with $p \in \mathbb{Z}$:

$$k(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^T \mathbf{x}')^p$$

which corresponds to a kernel obtained by a set of polynomial basis

functions:

$$k(\mathbf{x},\mathbf{x}') = (\mathbf{x}\cdot\mathbf{x}')^p = \left(\sum_{d=1}^{D} x_d x'_d\right)^p = \left(\sum_{d_1=1}^{D} x_{d_1} x'_{d_1}\right)\dots\left(\sum_{d_p=1}^{D} x_{d_p} x'_{d_p}\right) =$$
$$= \sum_{d_1=1}^{D}\cdots\sum_{d_p=1}^{D}(x_{d_1}\dots x_{d_p})(x'_{d_1}\dots x'_{d_p}) := \phi(\mathbf{x})\phi(\mathbf{x}')$$

Where the basis functions ϕ_m are given by all the monomials of degree p:

$$\phi_m(\mathbf{x}) = \sqrt{\frac{p!}{m_1!\dots m_D!}}x_1^{m_1}\dots x_D^{m_D}$$

with $\sum_j m_j = p$.

The **Rational Quadratic kernel** is defined by:

$$k_{QR}(r) = \left(1 + \frac{r^2}{2\alpha\ell^2}\right)^{-\alpha}$$

with $\alpha, \ell > 0$, which essentially is a scale mixture (an infinite sum) of squared exponential functions with different characteristic length-scales.

The **Periodic Kernel** (which allows us to embed periodic effects in the description of our process) is given by:

$$k(x,x') = \exp\left(-\frac{2\sin^2(\frac{x-x'}{2})}{\ell^2}\right)$$

where the 1-dimensional input variable x is mapped to the 2-dimensional $\mathbf{u}(x) = (\cos(x), \sin(x))$ to give rise to a periodic random function of x.

There are rules for constructing complex kernels starting from simpler ones. Given valid kernels $k_1(\mathbf{x},\mathbf{x}'), k_2(\mathbf{x},\mathbf{x}')$ the following kernels will also be valid:

$k(\mathbf{x},\mathbf{x}') = c\cdot k_1(\mathbf{x},\mathbf{x}')$ with $c > 0$
$k(\mathbf{x},\mathbf{x}') = f(\mathbf{x})k_1(\mathbf{x},\mathbf{x}')f(\mathbf{x}')$ with $f(\cdot)$ any function
$k(\mathbf{x},\mathbf{x}') = q(k_1(\mathbf{x},\mathbf{x}'))$ with $q(\cdot)$ polynomial with non-negative coefficients
$k(\mathbf{x},\mathbf{x}') = \exp(k_1(\mathbf{x},\mathbf{x}'))$
$k(\mathbf{x},\mathbf{x}') = k_1(\mathbf{x},\mathbf{x}') + k_2(\mathbf{x},\mathbf{x}')$
$k(\mathbf{x},\mathbf{x}') = k_1(\mathbf{x},\mathbf{x}')k_2(\mathbf{x},\mathbf{x}')$
$k(\mathbf{x},\mathbf{x}') = k_3(\phi(\mathbf{x}),\phi(\mathbf{x}'))$ with $\phi(\mathbf{x}) \in \mathbb{R}^M$, k_3 valid kernel
$k(\mathbf{x},\mathbf{x}') = \mathbf{x}^T A\mathbf{x}'$ with A symmetric poisitve semidefinite
$k(\mathbf{x},\mathbf{x}') = k_a(\mathbf{x}_a,\mathbf{x}'_a) + k_b(\mathbf{x}_b,\mathbf{x}'_b)$ with $\mathbf{x}_a, \mathbf{x}_b$ vars. s.t. $\mathbf{x} = (\mathbf{x}_a, \mathbf{x}_b)$ and k_a, k_b valid kernels
$k(\mathbf{x},\mathbf{x}') = k_a(\mathbf{x}_a,\mathbf{x}'_a)k_b(\mathbf{x}_b,\mathbf{x}'_b)$ as above

Combining different kernels allows us to combine multiple effects to describe our process at hand.

12.7 Hyperparameters optimization

Hyperparametrs of the kernel function have an impact on the covariance of the process. In practice, rather than fixing the covariance function, we may prefer to use a parametric family of functions and then infer the parameter values from the data.

The standard way to do so in a Bayesian approach is optimizing the (log) marginal likelihood. This is a data-driven method and works well if we can identify an analytical form for the log marginal likelihood. It turns out that this is our case:

$$\mathscr{L} = p(\mathbf{y}|\mathbf{x}) = \log \int p(\mathbf{f}|\mathbf{x})p(\mathbf{y}|\mathbf{f},\mathbf{x})d\mathbf{f}$$

We can rewrite it observing that $\mathbf{y} \sim \mathcal{N}(0, (K + \sigma^2\mathbb{I}))$:

$$\mathscr{L} = -\frac{1}{2}\mathbf{y}^T(K + \sigma^2\mathbb{I})^{-1}\mathbf{y} - \frac{1}{2}\log|(K + \sigma^2\mathbb{I})| - \frac{N}{2}\log 2\pi$$

The previous equation decomposes as:

- $-\frac{1}{2}\mathbf{y}^T(K + \sigma^2\mathbb{I})^{-1}\mathbf{y}$ which is the data fit;
- $-\frac{1}{2}\log|(K + \sigma^2\mathbb{I})|$ which is a complexity penalty term;
- $-\frac{N}{2}\log 2\pi$ which is a constant term.

Hence the marginal likelihood balances fit to the data and complexity penalty.

It is worth noting that the log marginal likelihood may be multimodal, hence not straightforward to optimize.

In order to maximize the marginal likelihood we can compute its derivative w.r.t. an hyperparameter θ, which is:

$$\begin{aligned}\frac{\partial}{\partial\theta_j}\log p(\mathbf{y}|\mathbf{x},\theta) &= \frac{1}{2}\mathbf{y}^T K^{-1}\frac{\partial K}{\partial\theta_j}K^{-1}y - \frac{1}{2}tr\left(K^{-1}\frac{\partial K}{\partial\theta_j}\right) = \\ &= \frac{1}{2}tr\left((\alpha\alpha^T - K^{-1})\frac{\partial K}{\partial\theta_j}\right)\end{aligned}$$

where $\alpha = K^{-1}\mathbf{y}$. Once we have evaluated K^{-1} (which we need also for the inference task), then it is easy to optimize $\mathscr{L}$.

We can also be fully Bayesian and place a prior on hyperparameters and find their posterior. If we do this, then we lose the analytical form. Hence with GP we typically work with marginal likelihood optimization to find hyperparameters.

As already mentioned, the typical choice for the prior mean is the zero function. In this case we usually realign the mean of the GP and the mean of the data by subtracting the sample mean from the data.

As an alternative, we can use a generalized linear model for the prior mean:

$$g(\mathbf{x}) = f(\mathbf{x}) + \mathbf{h}(\mathbf{x})^T \beta$$

where $f(\mathbf{x}) \sim GP(0, k(\mathbf{x}, \mathbf{x}'))$.

We can also be Bayesian and place a Gaussian prior on the weights $\beta \sim \mathcal{N}(\mathbf{b}, B)$ and get a GP:

$$g(\mathbf{x}) \sim GP(\mathbf{h}(\mathbf{x})^T \mathbf{b}, k(\mathbf{x}, \mathbf{x}') + \mathbf{h}(\mathbf{x})^T B \mathbf{h}(\mathbf{x}'))$$

In this way we obtain the a predictive distribution at a point $\mathbf{x}^\star$ having as mean and covariance respectively:

$$\bar{\mathbf{g}}(\mathbf{x}^\star) = H_\star^T \bar{\beta} + K_\star^T K_y^{-1}(\mathbf{y} - H^T \bar{\beta}) = \bar{\mathbf{f}}(\mathbf{x}^\star) + R^T \bar{\beta}$$
$$cov(\mathbf{g}^\star) = cov(\mathbf{f}^\star) + R^T (B^{-1} + H K_y^{-1} H^T)^{-1} R$$

where the H matrix collects the $\mathbf{h}(\mathbf{x})$ for all training (and $H_\star$ for all test) points, $\bar{\beta} = (B^{-1} + H K_y^{-1} H^T)^{-1}(H K_y^{-1} \mathbf{y} + B^{-1} \mathbf{b})$ and $R = H_\star H K_y^{-1} K_\star$.

Hence the mean of the new predictive distribution is $H_\star^T \bar{\beta}$ (from the linear model), plus a term coming from the GP model of residuals.

12.8 GP classification

In what follows we will consider the 2-class classification problem (everything can be easily extended to the multiclass classification).

Our goal is to model $p(C_1|x)$: we can take the Sigmoid function, so that the model is $p(C_1|x) = \sigma(f(x)) = \pi(x)$ with $f \sim GP(\mu, K)$.

We have observations of the form (x_n, y_n) with $y_n \in \{0, 1\}$ (hence our dataset is $(\mathbf{x}, \mathbf{y})$, $\mathbf{f} = f(\mathbf{x}) = f(x_1), \ldots f(x_N)$) and likelihood model $p(\mathbf{y}|f(\mathbf{x})) = \text{Bernoulli}(\sigma(f(\mathbf{x})))$ (as in logistic regression).

In this setting the function f is called *latent* (or nuisance) function (we don't directly observe it, but it governs the distribution of our data).

Remark: $\pi(\mathbf{x}) = \sigma(f(\mathbf{x}))$ is a random function.

The first thing that we want to evaluate is the probability of the latent function given the observations, by Bayes' theorem:

$$p(f(\mathbf{x})|\mathbf{y}) = \frac{p(\mathbf{y}|f(\mathbf{x}))p(f(\mathbf{x}))}{p(\mathbf{y})}$$

If we want to make a prediction at point $\mathbf{x}^\star$, first we need to compute the posterior at prediction points $\mathbf{x}^\star$:

$$p(f(\mathbf{x}^\star)|\mathbf{y}) = \int p(f(\mathbf{x}^\star)|f(\mathbf{x}))p(f(\mathbf{x})|\mathbf{y}) df(\mathbf{x})$$

Once we know this, we can compute the predictive distribution at $\mathbf{x}^\star$:

$$\pi^\star = \int \sigma(f(\mathbf{x}^\star))p(f(\mathbf{x}^\star)|\mathbf{y})df(\mathbf{x}^\star)$$

The problem is that, since the likelihood is a product of Bernoulli, there is no conjugacy (as in the case of Bayesian logistic regression), hence the posterior is not analytically tractable. The simplest solution is to rely on the Laplace approximation of the posterior $p(\mathbf{f}|\mathbf{y})$ around the MAP $\hat{f}$.

In order to do so we need to optimize the unnormalized log posterior, which reads as:

$$\begin{aligned}\Psi(\mathbf{f}) &= \log p(\mathbf{y}|\mathbf{f}) + \log p(\mathbf{f}|\mathbf{x}) = \\ &= \log p(\mathbf{y}|\mathbf{f}) - \frac{1}{2}\mathbf{f}^T K^{-1}\mathbf{f} - \frac{1}{2}\log|K| - \frac{n}{2}\log 2\pi\end{aligned}$$

Differentiating w.r.t. $\mathbf{f}$ we get:

$$\begin{aligned}\nabla\Psi(\mathbf{f}) &= \nabla \log p(\mathbf{y}|\mathbf{f}) - K^{-1}\mathbf{f} \\ \nabla\nabla\Psi(\mathbf{f}) &= \nabla\nabla \log p(\mathbf{y}|\mathbf{f}) - K^{-1} = -W - K^{-1}\end{aligned}$$

where W is diagonal, since observations are i.i.d.

We can optimize the log posterior via the Newton-Rapson scheme:

$$\begin{aligned}\mathbf{f}^{\text{new}} = \mathbf{f} - (\nabla\nabla\Psi)^{-1}\nabla\Psi &= \mathbf{f} + (K^{-1} + W)^{-1}(\nabla \log p(\mathbf{y}|\mathbf{f}) - K^{-1}\mathbf{f}) = \\ &= (K^{-1} + W)^{-1}(W\mathbf{f} + \nabla \log p(\mathbf{y}|\mathbf{f}))\end{aligned}$$

Plugging the definition of Laplace approximation around the MAP $\hat{f}$, we obtain a Gaussian q with mean:

$$\mathbb{E}_q[\mathbf{f}^\star|\mathbf{x}, \mathbf{y}, \mathbf{x}^\star] = k(\mathbf{x}^\star)^T K^{-1}\hat{f} = k(\mathbf{x}^\star)^T \nabla \log p(\mathbf{y}|\hat{f})$$

and variance:

$$\begin{aligned}\mathbb{V}_q[\mathbf{f}^\star|\mathbf{x}, \mathbf{y}, \mathbf{x}^\star] &= k(\mathbf{x}^\star, \mathbf{x}^\star) - k_\star^T K^{-1} k_\star + k_\star^T K^{-1}(K^{-1} + W)^{-1}K^{-1}k_\star = \\ &= k(\mathbf{x}^\star, \mathbf{x}^\star) - k_\star^T (K + W^{-1})^{-1} k_\star\end{aligned}$$

So that the prediction $\pi^\star$ can be computed by the integral:

$$\pi^\star \simeq \mathbb{E}_q[\pi^\star|\mathbf{x}, \mathbf{y}, \mathbf{x}^\star] = \int \sigma(\mathbf{f}^\star)q(\mathbf{f}^\star|\mathbf{x}, \mathbf{y}, \mathbf{x}^\star)d\mathbf{f}^\star$$

which can be approximated with the same logit-probit-logit trick used for Bayesian logistic regression.

However for GP typically we use different methods of approximation, belonging to the family of Variational Inference methods.

A first option is to approximate the posterior by a Gaussian q, which minimizes the KL divergence between q and the posterior distribution, i.e. $KL[q(\mathbf{f}|\mathbf{x}, \mathbf{y})||p(\mathbf{f}|\mathbf{x}, \mathbf{y})]$.

Alternatively we can use **Expectation Propagation**, which constructs iteratively a Gaussian approximation of the posterior, iterating over observations the following operations (observe that the posterior depends proportionally on the product of the likelihoods):

- take the current Gaussian approximation and factor out the term for the i-th likelihood $p(y_i|f_i)$, obtaining a distribution for all observations but the i-th one (called the cavity);
- multiply the cavity by the exact likelihood of the i-th observation, and find a Gaussian approximation by moment matching of such a (non-Gaussian) distribution.

This method converges faster than Laplace approximation typically (there is no optimization to run) and provides also an approximation of the marginal likelihood.

As a final observation, note that the main problems of GP prediction are:

- it is vulnerable to outliers, since adding a new observation always reduce the uncertainty at all points;
- optimization of hyperparameters is tricky, especially if σ^2 is unknown;
- GP predictions rely on a matrix inversion which scales cubically with the number of points (i.e. GP don't scale well with big data).